Question	Answer	Marks	Partial Marks
4(c)	11.7 or 11.66 … or 11.67	**3**	**M2dep** for $(4(their 2.5-1))^2+(2\times their 2.5+5)^2$ or **M1dep** for $4(their 2.5-1)$ or $2\times their 2.5+5$ OR **B1** for $\sqrt{20x^2-12x+41}$ and **M1dep** for substituting x = *their* 2.5 into $\sqrt{20x^2-12x+41}$ at any stage
5(a)	–3, 17	**2**	**B1** for each
5(b)	Fully correct curve	**4**	**B3 FT** for 10 or 11 points or **B2 FT** for 8 or 9 points or **B1 FT** for 6 or 7 points
5(c)(i)	Correct ruled tangent for *their* curve through (0, −17)	**1**	
5(c)(ii)	(1.7 to 2.2, –1 to 2.5)	**1**	
5(c)(iii)	[y =] $9x$ – 17 final answer	**3**	**M2dep** for answer [y =] $9x$[+] – c OR **M1dep** for gradient = $\frac{rise}{run}$ for *their* tangent at any point **B1** for answer [y =] kx[+] – 17 ($k \neq 0$)
5(d)	$y = 3x + 2$ ruled correctly and –2.2 … to –2.1 –0.6 to –0.4 2.6 to 2.8	**4**	**B2** for $y = 3x + 2$ ruled or **B1** for [y =] $3x + 2$ soi or $y = 3x + k$ ruled or $y = kx + 2$ but not $y = 2$ **B2** for all 3 values or **B1** for 2 values
6(a)	0.6	**1**	
6(b)	50.7	**3**	**M2** for $1.2 \times 19 + \frac{1}{2}(19 + 12) \times 1.8$ oe or **M1** for method for finding any relevant area
6(c)	17.9	**3**	**M2** for *their* 50.7 – 1.2 × 19 [– 10] oe or **M1** for 1.2 × 19 oe seen isw
7(a)	29	**1**	
7(b)	128	**2**	**FT** 180 – 2 (55 – *their* **(a)**) **M1** for angle *OCA* or angle *OAC* = 55 – *their* **(a)** soi

Question	Answer	Marks	Partial Marks
7(c)	64	**1**	**FT** *their* **(b)** ÷ 2
7(d)	116	**1**	**FT** 180 – *their* **(c)**
8(a)	370 or 370.2 to 370.3	**2**	**M1** for 864 ÷ *their* time
8(b)	991 or 990.5 …	**4**	**M2** for $864^2 + 928^2 - 2 \times 864 \times 928\cos 67$ or **M1** for correct implicit version **A1** for 981100 to 981110
8(c)(i)	313	**2**	**M1** for 180 + 133 or 360 – 47
8(c)(ii)	[0]79.5 to [0]79.6 …	**4**	**M2** for $\frac{928 \times \sin 67}{\textit{their } 991}$ or $\frac{864 \times \sin 67}{\textit{their } 991}$ oe or **M1** for implicit form of either **A1** for [angle *HGB* =] 59.5 to 59.6 … or [angle *HBG* =] 53.4 or 53.37 to 53.42 **M1 dep** for *their* angle *HGB* + 20 leading to answer or for 133 – *their* angle *HBG* leading to answer
9(a)(i)	42.8 or 42.79 … nfww	**4**	**M1** for mid-values soi **M1** for Σ*fm* where *m* is any value in interval including boundaries **M1** (dep on second **M1**) for *their* Σ*fm* ÷ 120
9(a)(ii)	Blocks of height 1.8 4.4 8 2.1 with correct widths	**4**	**B1** for each correct block If **B0**, **SC1** for correct frequency densities seen
9(b)	Valid general comment about distributions	**1**	e.g. [On average], shoppers spend less time shopping on Wednesday oe
10(a)(i)	75 000 × 60 × 20 oe	**M1**	Allow × 1200 for × 60 × 20
10(a)(ii)	16.4 or 16.36 …	**3**	**M2** for $\frac{9 \times 10^7 \times 100}{1000 \times 55 \times 10^4}$ oe or **B2** for answer 0.164 or 0.1636 … or **B1** for answer figs 164 or 1636 … or **M1** for figs 9 ÷ figs 55
10(a)(iii)	28.3 or 28.27 to 28.28	**3**	**M2** for $\frac{76}{360} \times 2\pi \times 8.5 + 2 \times 8.5$ oe or **M1** for $\frac{76}{360} \times 2\pi \times 8.5$ oe